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DETAIL SPECIFICATION

LACE, GOLD – ORNAMENTAL

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for synthetic, tarnish-resistant ornamental gold lace with an interwoven scarlet center stripe.

1.2 Classification. The lace will be of one type in the following classes, as specified (see 6.2).

Class 1 – 9.525 mm (3/8 inch) wide

Class 2 – 18.875 mm (5/8 inch) wide

Class 3 – 31.75 mm (1-1/4 inches) wide

Class 4 – 38.1 mm (1-1/2 inches) wide

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-E-17568 - Embroidery Materials, Metallic and Synthetic Metallic

(Copies of this document are available online at https://quicksearch.dla.mil/.)

Comments, suggestions, or questions on this document should be addressed to Marine Corps Systems Command, 2200 Lester Street, Quantico, VA 22134 ATTN: SEAL-SE-STDS or emailed to USMC_STDZ@usmc.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

16 CFR 303 - Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at www.ecfr.gov.)

THE INSTITUTE OF HERALDRY (TIOH)

Department of Defense Standard Shades for Heraldic Yarns for Embroidery and Ribbons

(Copies of the chip set are available from The Institute of Heraldry, 9325 Gunston Road, Room S-113, Fort Belvoir, VA 22060-5579, or by emailing us.army.belvoir.hdqa.mail.tioh-webmaster@mail.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1 - Gray Scale for Color Change

AATCC EP2 - Gray Scale for Staining

AATCC EP8 - AATCC 9-Step Chromatic Transference Scale

AATCC EP9 - Visual Assessment of Color Difference of Textiles

AATCC TM8 - Colorfastness to Crocking: Crockmeter

AATCC TM15 - Colorfastness to Perspiration

AATCC TM16.3 - Test Method for Colorfastness to Light: Xenon-Arc

AATCC TM20 - Test Method for Fiber Analysis: Qualitative

AATCC TM132 - Colorfastness to Drycleaning

AATCC TM133 - Colorfastness to Heat: Hot Pressing

(Copies of these documents are available online at www.aatcc.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance inspection. When specified (see 6.2), a sample shall be subjected to a conformance inspection in accordance with 4.3.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Materials.

3.3.1 Gold synthetic metallic yarn. The gold synthetic metallic embroidery yarn shall conform to type III, class 8 or class 11 of MIL-E-17568. The core yarn may be silk, nylon, or cotton. The gold synthetic metallic yarn shall meet the color and colorfastness requirements as specified in 3.4.1.

3.3.2 Center stripe. The scarlet center stripe of lace shall be interwoven (warpwise direction) with silk thread.

3.4 Lace design and construction. The lace shall be designed and constructed to meet the physical requirements listed in table I when tested as specified in 4.5.

TABLE I. Design and construction requirements.

Class	Width (min.)	Weave, pattern	Picks (min.)	Ribs (min.)	Pattern width		
					Gold	Center stripe	Gold
1	9.525 mm (3/8")	2 per cm (5 per inch)	102±1 per cm (260 per inch)	13.5±0.5 per cm (35 per inch)	4.36 mm (11/64")	0.8 mm (1/32")	4.36 mm (11/64")
2	18.875 mm (5/8")	2 per cm (5 per inch)	102±1 per cm (260 per inch)	13.5±0.5 per cm (35 per inch)	7.14 mm (9/32")	1.6 mm (1/16")	7.14 mm (9/32")
3	31.75 mm (1-1/4")	2 per cm (5 per inch)	102±1 per cm (260 per inch)	8.2±0.25 per cm (21 per inch)	12.7 mm (1/2")	6.35 mm (1/4")	12.7 mm (1/2")
4	38.1 mm (1-1/2")	2 per cm (5 per inch)	102±1 per cm (260 per inch)	8.2±0.25 per cm (21 per inch)	16.7 mm (21/32")	4.76 mm (3/16")	16.7 mm (21/32")

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3.4.1 Lace requirements. The lace shall conform to the requirements listed in table II when tested as specified in 4.5. All materials used shall be capable of withstanding dry cleaning without significant degradation.

TABLE II. Lace colorfastness requirements.

Characteristic	Requirement
Color:	
Core yarn and synthetic metallic ribbon	Gold Shade, Cable no. 67104
Center stripe thread	Scarlet, Shade W, Cable no. 67111
Colorfastness:	
Crocking (wet and dry)	3 to 5
Perspiration (acid and alkaline) (color change and staining)	4
Light (40 AFU 1/ or 170 kJ/[m^2nm] 2/ at 420 nm)	4
Drycleaning (color change and staining)	4
Hot pressing (wet and dry) at 150 °C (color change and staining)	4
Labile sulfur (when cotton fiber is used)	"Slight" or "Free"
FOOTNOTES: 1/ AATCC Fading Units (AFU) 2/ Kilojoules (kJ) per square meter per nanometer (m^2nm).	

3.5 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the lace shall be put on rolls that are 22.8 meters (25 yards) in length total. One splice per roll shall be allowed, but no spliced piece shall be less than 4.6 meters (5 yards) in length. The lace shall be wound on suitable cores of the same width as the lace. The ends of the rolls shall be securely fastened to prevent slippage and unrolling of the roll.

3.6 Identification. Each roll shall have an identification ticket attached. The following information shall be legibly printed on the ticket with water insoluble ink:

a. Nomenclature
b. Specification number
c. Length
d. Contract number and date
e. Supplier's name
f. Name of contracting agency

3.6.1 Fiber identification. Each roll of lace shall be labeled or ticketed for fiber content in accordance with 16 CFR 303 when examined as specified in 4.4.2.

3.7 Workmanship. The finished lace shall conform to the quality of product established in this specification. The occurrence of defects shall not exceed the Acceptance Quality Limits (AQLs) as specified in the contract (see 6.2).

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. Unless otherwise specified (see 6.2), a first article, submitted in accordance with 3.1.1, shall be inspected, examined for appearance and finished defects listed in 4.4, and tested for the characteristics as specified in 4.5.

4.3 Conformance inspection. Unless otherwise specified (see 6.2), conformance inspection, in accordance with 3.1.2, shall include the visual examination of 4.4 and the tests of 4.5 through 4.6, as applicable.

4.3.1 Sampling for conformance inspection. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with AQLs as specified in the contract (see 6.2), except where otherwise indicated herein.

4.3.2 Inspections. Unless otherwise specified, excluded, amended, modified, or qualified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with all the requirements of referenced documents.

4.4 Meter-by-meter examination. Each roll of lace in the sample shall be examined meter-by-meter on the face side of the lace for defects. The defects are as defined in ASTM D3990, and only defects that are clearly noticeable at a normal inspection distance of 1 meter shall be scored. The defects are assigned demerit points in accordance with ASTM D5430, option A, with the exceptions listed below:

a. The defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the lace, in which case only the more serious defect shall be scored.

b. A continuous defect shall be counted as one defect for each warpwise meter or fraction thereof in which it occurs.

c. Only coarse yarns that exceed twice or more the normal diameter shall be scored.

4.4.1 Demerit point computation. The point computation shall be as follows:

$$\frac{\text{Total points scored in sample} \times 10{,}000}{\text{Width of cloth in centimeters} \times \text{total meters inspected}} = \text{Points per } 100\ \text{m}^2$$

4.4.2 Defects. The defects in table IV shall be scored four points for each meter in which they occur. No linear meter (increments of 1 meter on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than four points. The number of rolls to be examined shall be selected as specified in table III.

TABLE III. Sample size.

Lot size (in meters)	Sample size (in rolls)
Up to 2,926	10
2,927 to 9,144	15
9,145 to 32,004	25
35,005 and over	35

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TABLE IV. Defects.

#	Defect description
1	Any hole, cut, pinhole, tear, scratch, or abrasion mark
2	Broken or missing yarn (end or pick)
3	Uneven tension resulting in thick and thin places
4	Tight or loose yarn causing distortion or loopiness
5	Tight picks
6	Coarse or heavy picks
7	Fine or light picks
8	Variation of shade in same roll
9	Selvages curled, folded, or rolled
10	Any spot or stain
11	More than one splice

4.4.3 Length examination. During the meter-by meter examination, each roll of lace in the sample shall be examined for length. The unit of product for this examination shall be one roll. The sample size shall be expressed in units of one roll. The inspection level shall be as specified in the contract (see 6.2).

4.4.3.1 Individual pieces. The pieces shall be examined for gross length. Any gross length found to be less than the specified minimum length, any gross length found to be more than 1 meter below the gross length marked on the piece ticket, or any spliced piece less than 5 meters long shall be considered a defect with respect to length.

4.4.3.2 Total yardage in sample. The lot shall be unacceptable if the actual total of the gross length of pieces in the sample is less than the total of the gross length marked on the piece tickets.

4.4.4 Roll identification examination. During the meter-by-meter examination, each roll in the sample shall be examined to ensure ticketing or labeling compliance with 16 CFR 303. The lot shall be unacceptable if two or more rolls are not in compliance.

4.5 Physical requirements testing. The lace shall be tested for the characteristics listed in table V. The methods of testing as specified where applicable herein and as listed in table V shall be followed. As specified (see 6.2), all test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if one or more tests fail to meet the requirement specified. The inspection level shall be as specified in the contract (see 6.2).

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TABLE V. Physical requirements thread testing.

Characteristic	Requirement	Test method
Fiber identification		
Nylon fiber	3.3	AATCC TM20 1/
Silk	3.3	AATCC TM20 1/
Cotton	3.3	AATCC TM20 1/
Lace requirements		
Width in millimeters	Table I	Visual 2/
Fancy weave pattern	Table I	Visual 2/
Picks per centimeter	Table I	Visual 2/
Ribs per centimeter	Table I	Visual 2/
Width of pattern	Table I	Visual 2/
Color (visual shade matching)	Table II	AATCC EP9, opt. C (see 6.3)
Daylight D65 (6500±200 Kelvin [K]) (with illumination of 100±20 foot candles) (see 6.5)		
Incandescent (2856±200K)		
Colorfastness (minimum)		
Crocking	Table II	AATCC TM8
Wet and dry	Table II	AATCC EP8
Perspiration (acid and alkaline)	Table II	AATCC TM15
Color change	Table II	AATCC EP1
Staining	Table II	AATCC EP2
Light (after 40 hours or 170 kilojoules)	Table II	AATCC TM16.3, opt. 3
Color change	Table II	AATCC EP1
Dry cleaning (after 4 cycles)	Table II	AATCC TM132
Color change	Table II	AATCC EP1
Staining	Table II	AATCC EP2
Heat: hot pressing (wet and dry) at 150 °C	Table II	AATCC TM133
Color change	Table II	AATCC EP1
Staining	Table II	AATCC EP2
Labile sulfur	Table II	4.6

FOOTNOTES:

1/ See 6.4.

2/ One determination shall be made from each sample and results reported as “pass” or “fail”.

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4.6 Presence of labile sulfur test. The cloth shall meet the requirements specified in table II. In the determination of presence of labile sulfur in textile materials with lead acetate, two 1.50±0.01-gram samples from each material submitted for evaluation shall be tested. Each of the two samples shall be cut into very small pieces and placed into separate test tubes. The samples shall be submersed into a stannous chloride solution that contains 100 grams of American Chemical Society (ACS) grade stannous chloride crystals in 100 milliliters of ACS grade hydrochloric acid (35 percent concentration) and 50 milliliters of distilled water. A filter paper wet out with a 5.0 percent lead acetate solution shall be placed over the top of the test tube. The lead acetate solution contains 5.0 grams of chemically pure (CP) reagent grade lead acetate and enough distilled water to make up a 100-milliliter solution; if the solution is not clear, a few drops (one at a time) of glacial acetic acid shall be added until the solution is clear. The test tube containing the textile sample, stannous chloride, and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown or black stain on the filter paper shall be evaluated as follows:

a. Free – The filter paper shows no discoloration or staining of any kind.
b. Slight – The filter paper shows a light tan to light brown discoloration stain.
c. Moderate – The filter paper shows a dark brown discoloration stain.
d. Severe – The filter paper shows a black color stain.

Any rating other than “Free” or “Slight” shall be considered a test failure.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The gold lace covered by this specification is intended to be used on clothing and accessory items as follows:

a. Class 1 – Chin strap for women’s Marine dress caps.
b. Class 2 – Chin strap for men’s Marine dress caps.
c. Class 3 – Marine Corps Evening dress trousers.
d. Class 4 – Full dress belt for Director and Assistant Director, U.S. Marine Corps Band.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class required (see 1.2).
c. The specific issue of individual documents referenced (see 2.2 and 2.3).
d. When first article is required (see 3.1.1 and 4.2).
e. When conformance inspection is required (see 3.1.2 and 4.3).
f. Length put-up required (3.5).
g. AQLs required (see 3.7 and 4.3.1).
h. Inspection conditions (4.3.2).
i. Test reports documenting the individual values of physical requirements inspection (see 4.5).
j. Physical requirements inspection level (see 4.5).

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k. Packaging requirements (see 5.1).

l. Certificates of compliance, if acceptable for physical requirements (see 6.4).

6.3 Visual shade matching. In 2019, option A of AATCC EP9 was changed to option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle relative to the plane of the specimens".

6.4 Certificates of compliance. Certificates of compliance may be accepted for selected requirements (see 6.2). It is recommended that vendors verify acceptability prior to using certificates of compliance to address requirements.

6.5 Use of D75 illuminant. The use of D75 illuminant with a color temperature of 7500±200 Kelvin and illumination of 100±20 foot candles in lieu of the specified D65 illuminant (see table V) is permitted.

6.6 Subject term (key word) listing.

Center stripe

Chin straps

Trousers

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – MC

Review activity:
Navy – NU

Preparing activity:
Navy − MC
(Project 8315-2022-001)

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